

Microbial association networks

Introduction to Metagenomics

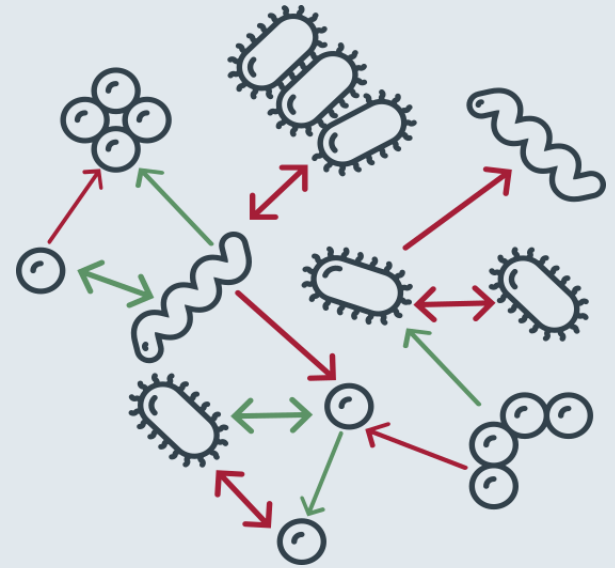
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Data Science Platform (DSP)

DTU-Biosustain

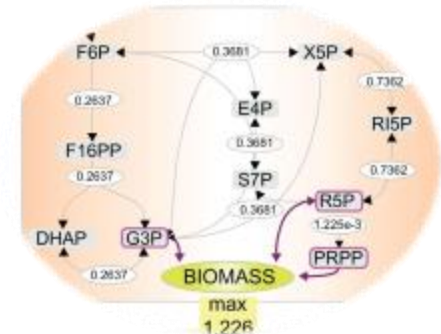
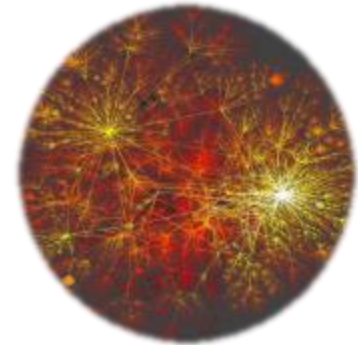
Spring Courses

June 11, 2025



What is a Network?

- Data structure of **components (nodes)** connected by **relationships (edges)**
- **Broad concept** that describes how things interact to each other
- Networks are everywhere



Why Networks?

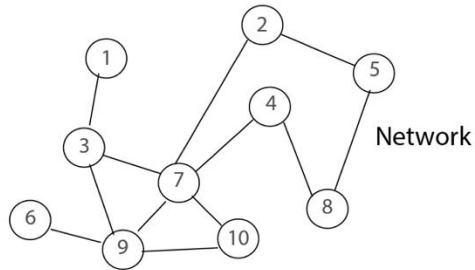
These structures allow:

- Quick **integration** of **heterogeneous data** based on relationships
- **Graph theory** methods can be used to **analyze** and **interpret** data, e.g., topological properties can be used to explain:
 - The possible **role** of specific components
 - The identification of **clusters** of nodes
 - The **flow** of information
 - The **robustness** of the system
- Examine **similarities and differences** between systems
- **Visualize** data – complex systems can be represented as networks



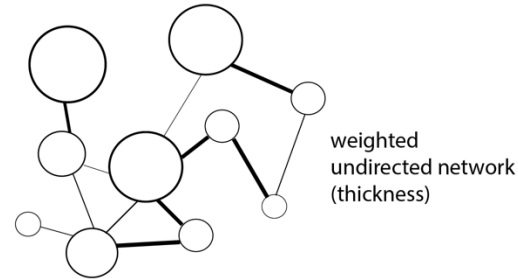
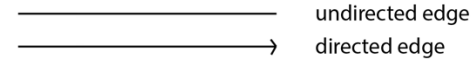
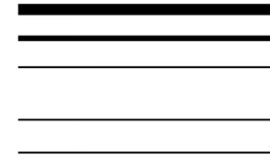
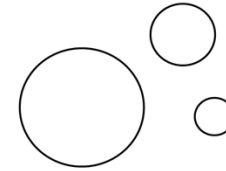
Basic Network Concepts

$$G = (V, E)$$



0	0	1	0	0	0	0	0	0	0
0	0	0	0	1	0	1	0	0	0
1	0	0	0	0	1	0	1	0	0
0	0	0	0	0	0	1	1	0	0
0	1	0	0	0	0	1	0	0	0
0	0	0	0	0	0	0	0	0	1
0	1	1	1	0	0	0	0	1	1
0	0	0	1	1	0	0	0	0	0
0	0	1	0	0	1	1	0	0	1
0	0	0	0	0	0	1	0	1	0

Adjacency matrix



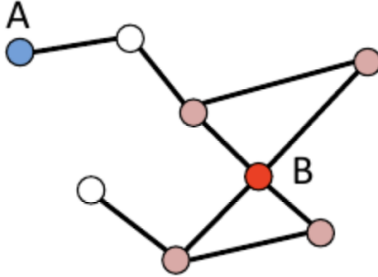
0	0	w ₁₂	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
w ₁₂	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	w ₁₂	w ₁₃	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	w ₁₂	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

Weighted adjacency matrix

Basic Network Concepts – Degree centrality

- **Centrality:** measure of how important a node is in the network

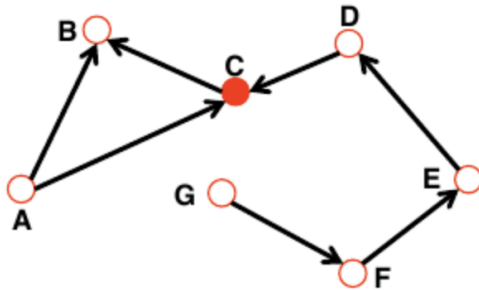
Undirected



Node degree: the number of links connected to the node.

$$k_A = 1 \quad k_B = 4$$

Directed



In *directed networks* we can define an **in-degree** and **out-degree**.

The (total) degree is the sum of in- and out-degree.

$$k_C^{in} = 2 \quad k_C^{out} = 1 \quad k_C = 3$$

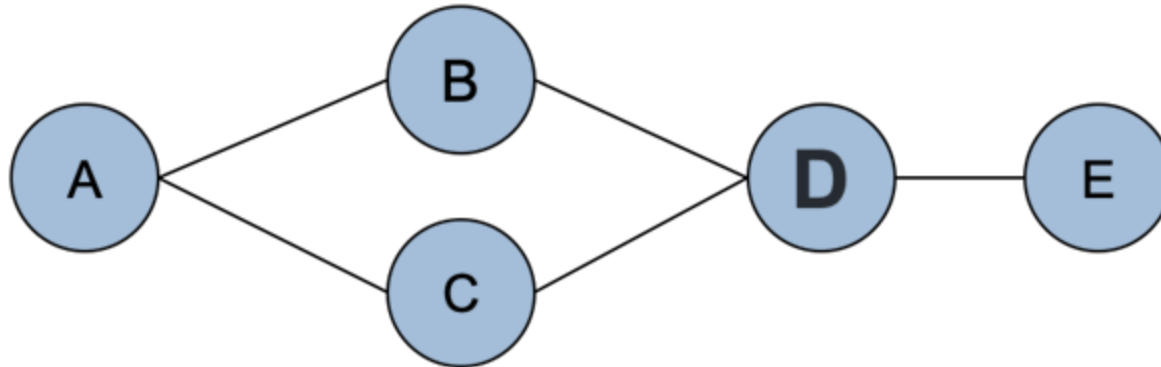
Basic Network Concepts – Betweenness centrality

- The number of times a node is **in between** an origin and a destination
- How many paths will become longer if a given node disappears from the network?
- Nodes with a **high betweenness** centrality can control **information flow**

$$b_i = \sum_{r,s} \frac{n_{rs}^i}{g_{rs}}$$

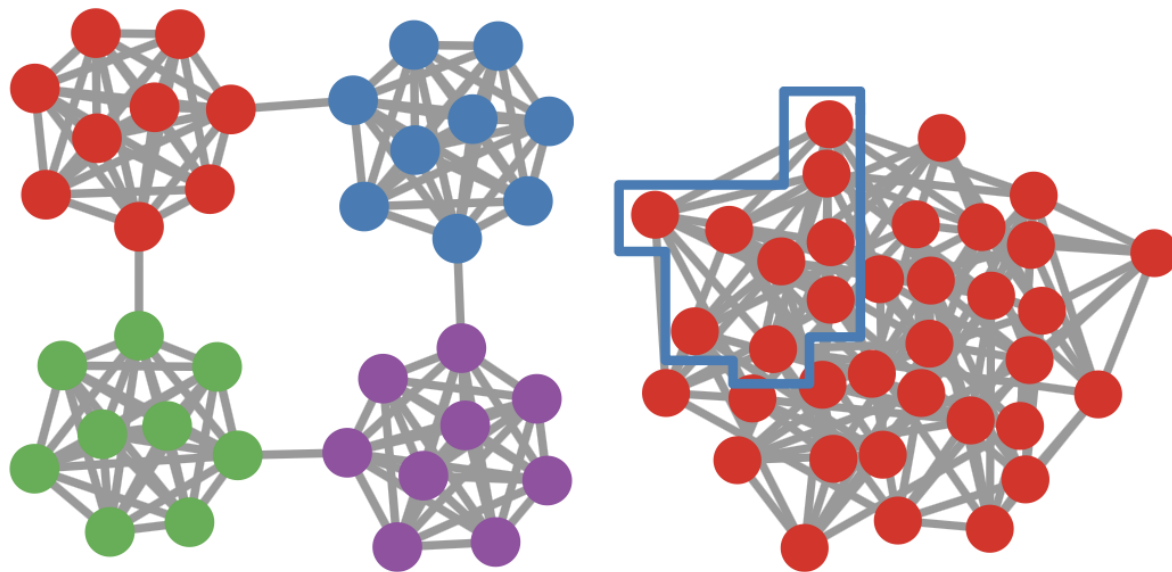
Number of shortest paths between r and s that pass through i

Number of shortest paths between r and s



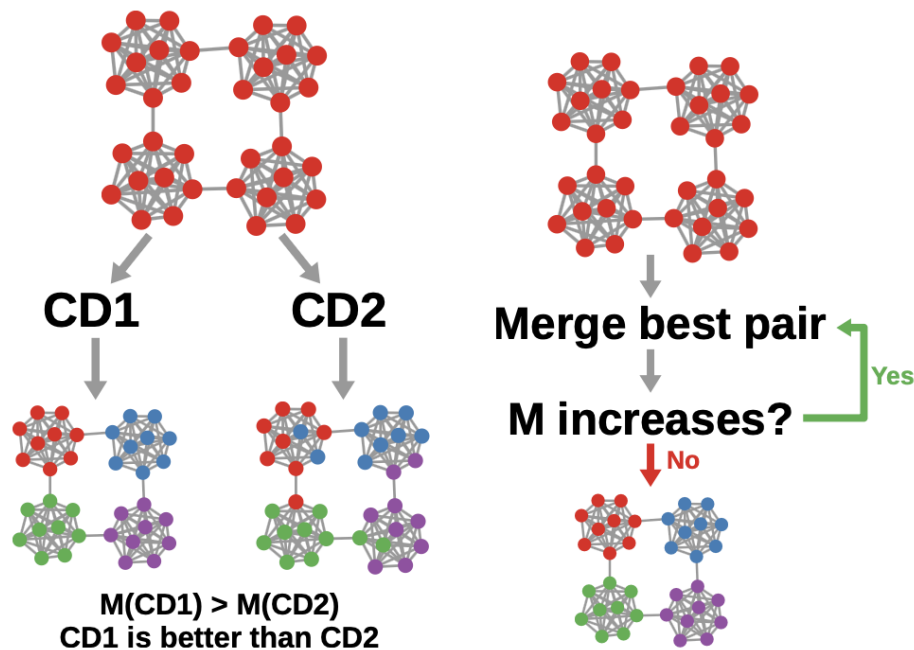
Basic Network Concepts - Clustering

- **Clustering:** finding groups of nodes that are **more connected to each other** than to the rest of the network
- **Modularity:** comparison between the **observed number of edges** inside a community and the **expected number of edges in a null graph** model with random edges



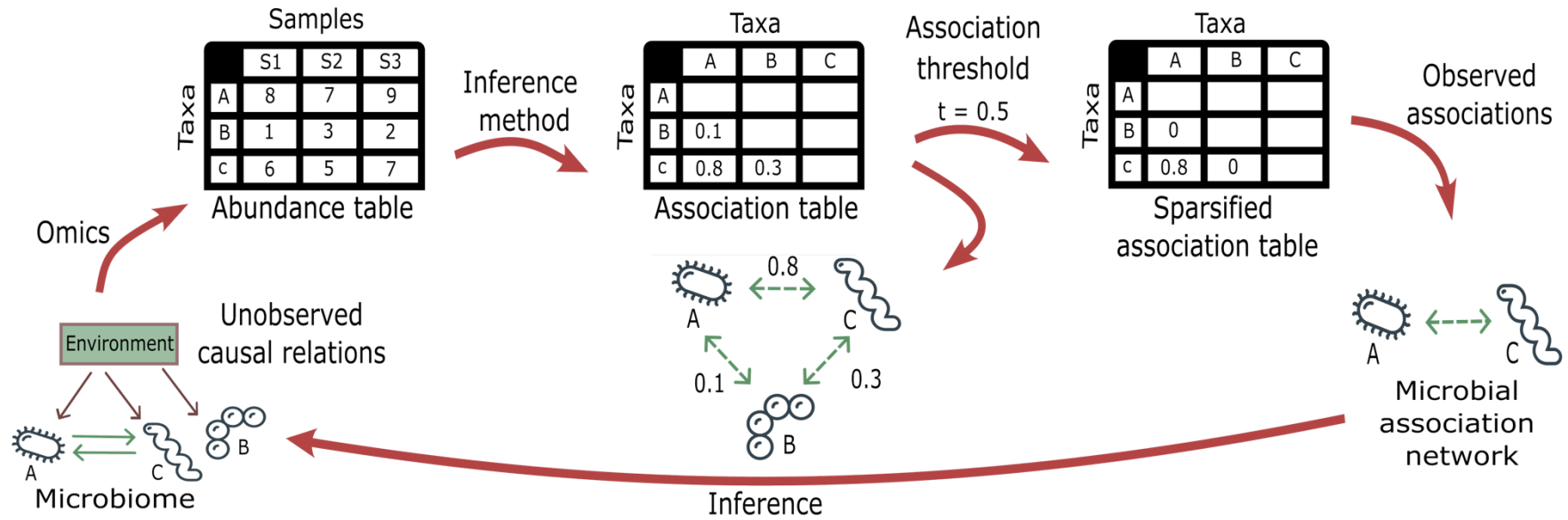
Basic Network Concepts – Clustering

- **Modularity** gives us an intuition about the **overall community structure** - higher modularity implies a **better partition**
- **Modularity optimization**: iteratively grouping nodes into communities, aiming to **maximize** the **intra-community edges** compared to inter-community edges



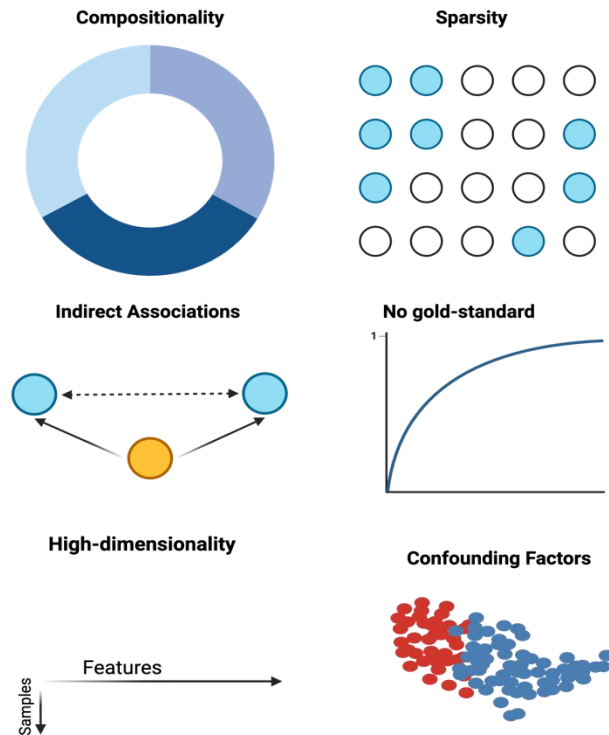
Microbial association networks (MANs)

- **MAN: weighted** and **undirected** network, where nodes are microbes and edges indicate **co-presence** or **mutual exclusion abundance patterns** across samples
- **Weights** are **association values** among microbes defined by an **inference method**



Limitations and Challenges of MANs

- **Compositionality of microbial data** — only **relative abundances** are measured, not absolute counts
- **Sparsity of data** — containing **many zero counts** due to low abundance or detection limits
- **High-dimensionality with small sample size** — Often the number of microbial taxa (**features**) greatly **exceeds** the number of samples (**observations**)
- **Indirect associations** — hard to **distinguish** actual **interactions** from **secondary effects**
- **Lack of ground truth** — no “gold standard” network to validate inferred MANs
- **Confounding factors** — **host factors** (e.g., diet, health, medications) or **technical artefacts** (e.g., batch effects, depth)



Do you want to know more?

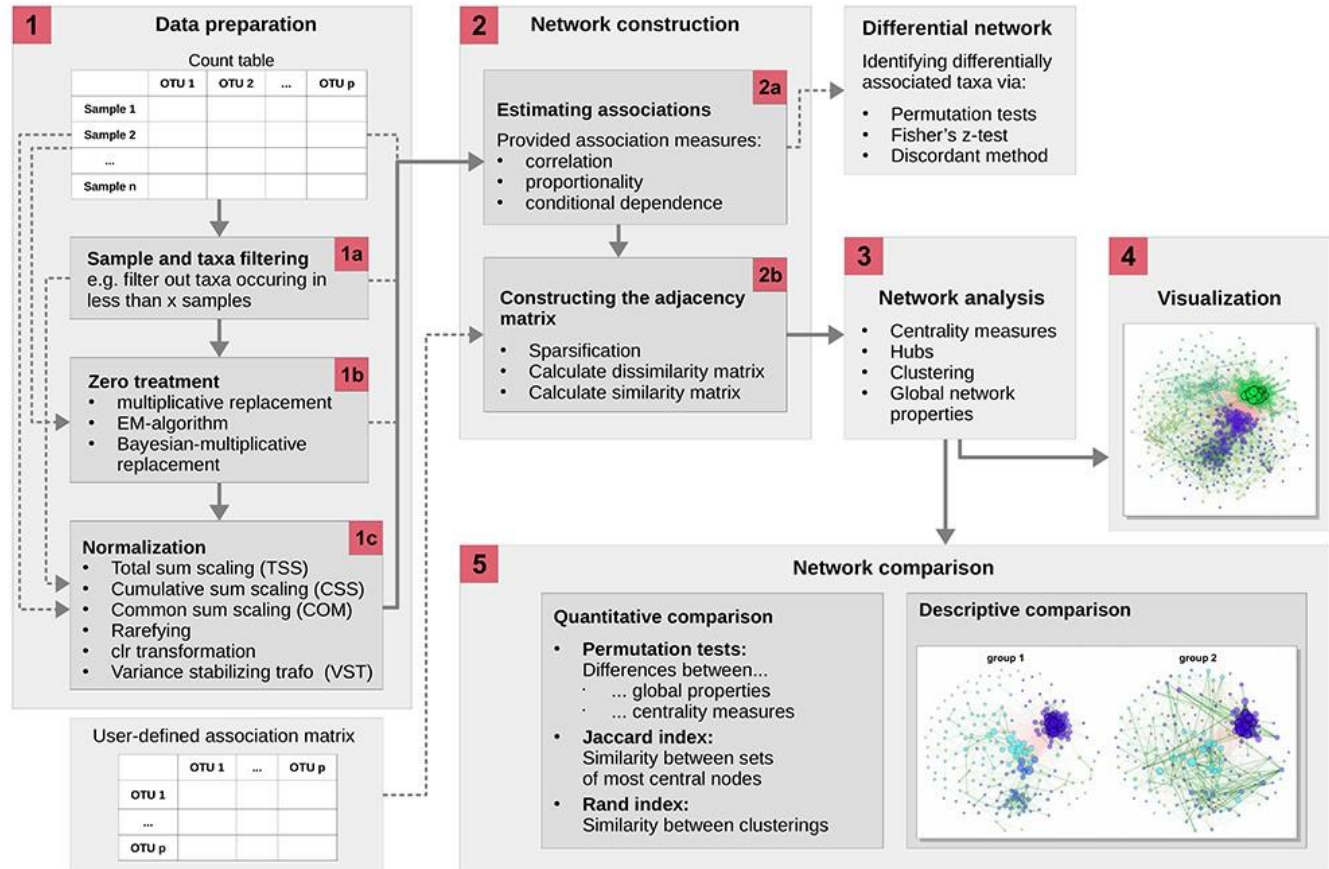
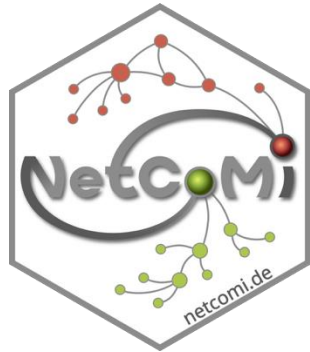
- **Course GitHub repo:** https://github.com/biosustain/dsp_metagenomics_training
- **Docs:** https://biosustain.github.io/dsp_metagenomics_training
- **NetCoMi:** <https://netcomi.de/>
- **MANs:** https://microbiome.github.io/OMA/docs/devel/pages/network_learning

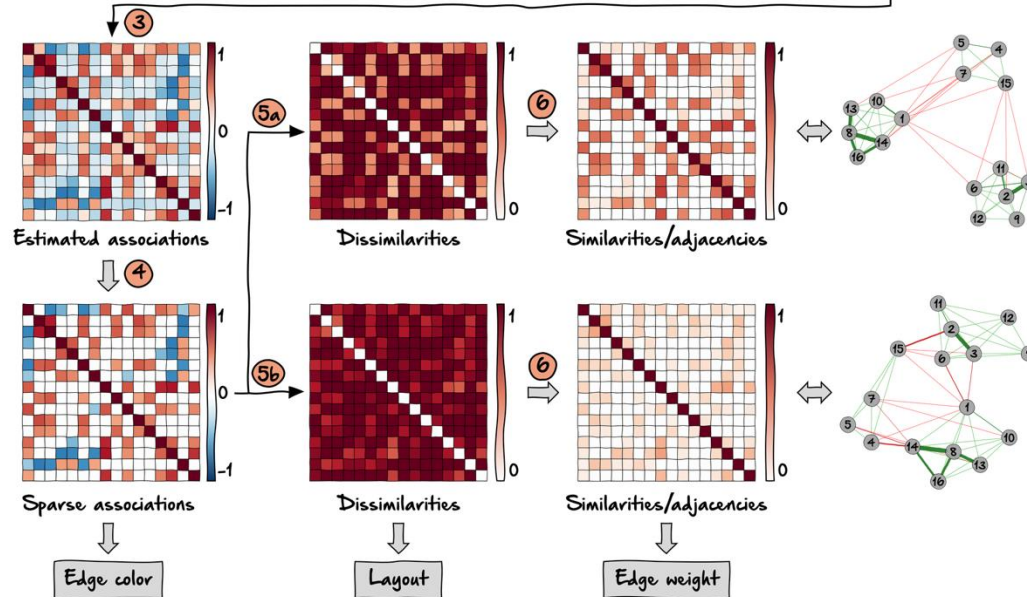
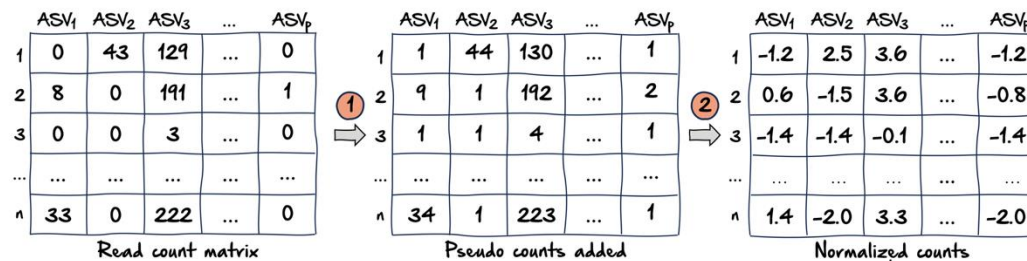
References

- Layeghifard, M., Hwang, D. M., & Guttman, D. S. (2017). Disentangling Interactions in the Microbiome: A Network Perspective. *Trends in Microbiology*, 25(3), 217–228
- Coscia, M. (2025). The atlas for the aspiring network scientist. Second Edition. arXiv: 2101.00863

Extra slides

NetCoMi – General Workflow





NetCoMi – Network Types

